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Satellite micro-launcher control: An integrated adaptive-robust and nonlinear approach

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ABSTRACT

In this paper, a composite and multi-technique-based controller for a three-stage micro-launcher is presented. The proposed control law consists in a linear robust component, obtained for the rigid motion using mixed-sensitivity and normalized coprime factor loopshaping, an adaptive command gain, used to compensate the effects of the flexible motion, and a nonlinear component, used for roll control and payload insertion. Moreover, robust stability is ensured in the presence of parametric uncertainty, while also guaranteeing a set of disk margins which accommodate the adaptive gain. Finally, robust performance is validated via structured singular value analysis at successive instants along the trajectory, while the control scheme is tested in simulation upon the micro-launcher's nonlinear and uncertain dynamics.

1. Introduction

With the demand for orbital insertion missions skyrocketing in recent years, both private and public sectors have turned towards finding more economical alternatives to classically-built satellite launchers (Du, 2010; Marcos et al., 2020). One prominent option in this class of economically-viable solutions is represented by the so-called "satellite micro-launchers", owing to their reduced size. Although showing great potential, the control solutions available in literature for this class of launchers center around augmenting classical control laws or obtaining low-order single-input single-output robust controllers (Bordeneuve-Guibé et al., 2014). Thus, under the aegis of the European Space Agency, a control solution was proposed for the prototype which has been mathematically modeled in Chelaru (2018), which integrates diverse modern control techniques (Blight et al., 1994; Papageorgiou & Glover, 1999).

In order to face the specific challenges posed by the considered micro-launcher, the approach presented in this paper had to overcome a number of design challenges. Firstly, the obtained control scheme had to be both straightforward and modular, while minimizing the need for gain scheduling and avoiding the need for complex interpolation techniques [see Clement et al. (2005)]. Note, however, that although gain scheduling is not difficult to accomplish in practice, the proposed solution significantly simplifies the process of formally verifying closed-loop stability. Secondly, the rigid-body dynamics of the micro-launcher had to be robustly stabilized, even as the sources of parametric uncertainty for the model described in Chelaru (2018) formed nonlinear combinations between themselves, rendering the direct LFT approach from Simplício et al. (2016) unsuitable for the considered application. To further compound this point, a preliminary check which consisted in grouping up the uncertain terms and performing a corner-case analysis (Valli et al., 2015) yielded results which proved far too conservative for the procedure given in Simplício et al. (2016) to ensure any significant performance. Moreover, the flexible-body's bending modes, atmospheric interference during the first phase of the ascent and the roll-channel cross-coupling with the pitch and yaw axes were shown to induce significant closed-loop disturbance, which can lead to failed payload insertion.

The chief contribution of this paper consists in designing a composite control scheme, whose constituent parts deal with each of the aforementioned challenges and whose concerted efforts ensure mission success. The main thrust vectoring controllers robustly stabilize the launch vehicle with respect to parametric uncertainty by employing its equivalent unstructured formulation, obtained by solving an optimization problem over an extensively sampled data set. Additionally, these controllers are obtained by exploiting the fact that the micro-launcher's nominal mathematical model does not change dramatically during any of its stages. Therefore, only two thrust vectoring controllers are employed during each of the three stages (one for the pitch axis

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Nomenclature

$\theta,\ \psi,\ \phi$	pitch, yaw and roll angles;
$\theta_r,\ \psi_r$	pitch and yaw reference angles;
$\delta_\theta,\ \delta_\psi$	pitch and yaw thrust vectoring control deflection angles;
$\delta_\theta^{AUG},\ \delta_\psi^{AUG}$	pitch and yaw thrust vectoring control augmented deflection angles;
$u_\theta,\ u_\psi,\ u_\phi$	pitch, yaw and roll on–off-reverse commands for the attitude and orbit control system;
$\dot{z}$	lateral drift speed;
$\alpha,\ \beta$	angle of attack and sideslip angle;
α_w	pitch axis wind incidence angle;
$C_{L\alpha}$	lift coefficient;
$C_{n\alpha}$	coefficient of the normal aerodynamic force slope with respect to the incidence;
C_x	moment of inertia on the roll axis;
C_{yz}	moment of inertia on both the pitch and the yaw axes;
m	mass;
T	thrust force;
ℓ	distance between the center of mass and the aerodynamic center of pressure;
V	magnitude of the velocity vector;
F_0	aerodynamic reference force;
H_0	aerodynamic reference moment;
R	distance between the center of mass and the center of the Earth;
θ_i	ith pitch axis bending mode;
K_θ	pitch axis bending constant;
$\theta_{tot},\ \psi_{tot}$	pitch and yaw angles while accounting for roll-induced coupling;
δ_{THR}	cold-gas thruster's thrust level;
u_{THR}^{ϕ}	cold-gas thruster input to be converted into an on–off-reverse signal;
τ_{THR}	time constant of thruster actuator model;
T_{THR}	thrust force of thruster actuator model.

and the other one for its yaw counterpart), thus eliminating the need for a complex gain-scheduling algorithm. These controllers are then weighted by an adaptive gain augmentation scheme, introduced in Orr et al. (2014), Orr and Zwieten (2012) [see also its alternative in Oh et al. (2008)] and further refined in Navarro-Tapia et al. (2021), which has the benefit of compensating the effect of launcher bending modes without the need to redesign or even retune the rigid-body controllers. Furthermore, the proposed nonlinear roll-control system minimizes roll-induced disturbance, freeing up the thrust vectoring controllers to focus on trajectory tracking and rejection of atmospheric perturbation. A norm-based validation and verification procedure is proposed to complement the nondeterministic nature of classical Monte-Carlo simulation campaigns. Finally, due to the prototypical nature of the model from Chelaru (2018), the control solution is implemented for all three of the micro-launcher's stages, from lift-off to payload insertion, to showcase its effectiveness.

The paper is organized as follows. Section 2 covers the modeling of the micro-launcher and its flight mission. Section 3 addresses control system design. Section 4 details the validation procedure for the obtained control laws. The control scheme is tested upon the micro-launcher's nonlinear flight simulator and numerical simulations are showcased in Section 5. Conclusions and future improvements are presented in Section 6.

2. Mission profile and micro-launcher model

2.1. Guidance algorithm

The fundamental requirement of the mission rests in placing the launch vehicle's payload in orbit around the Earth, while ensuring the orbit's eccentricity is minimized. In the case of the considered micro-launcher, the ascent lasts just under 21 minutes and is divided into two phases, shown in Figs. 1(a) and 1(b).

The first phase covers the atmospheric part of the ascent. For the first few seconds, while the micro-launcher clears the tower, its attitude is kept at the initial values. Following this, the launch vehicle pitches over from its vertical position at 2 deg/s for around 11 s, while preventing any yaw movement. Once this is done, the achieved attitude is preserved for a few more seconds, to allow for the alignment between the micro-launcher's velocity vector and its longitudinal axis. At the 17 second mark, the gravity turn maneuver begins and lasts until the end of the atmospheric segment, just after the 6 minute mark. The launcher's first separation occurs between 68 and 71 s, while the second, occurring between 123 and 142 s, marks the beginning of the exo-atmospheric phase.

Remark 1. To elaborate upon the claims pertaining to the ascent trajectory, refer to Fig. 1(a) in which both altitude and Mach number are displayed as functions of time, along with the two aforementioned coasting phases, marked in dashed lines with respect to the altitude axis, on the left-hand side of the plot. Note that coasting times may be inspected with respect to trajectory color, as depicted in the color gradient axes on the right-hand side of the plots. The coasting phases are more clearly shown in Fig. 1(b) and are distinguishable as the only segments where the micro-launcher's Mach number decreases.

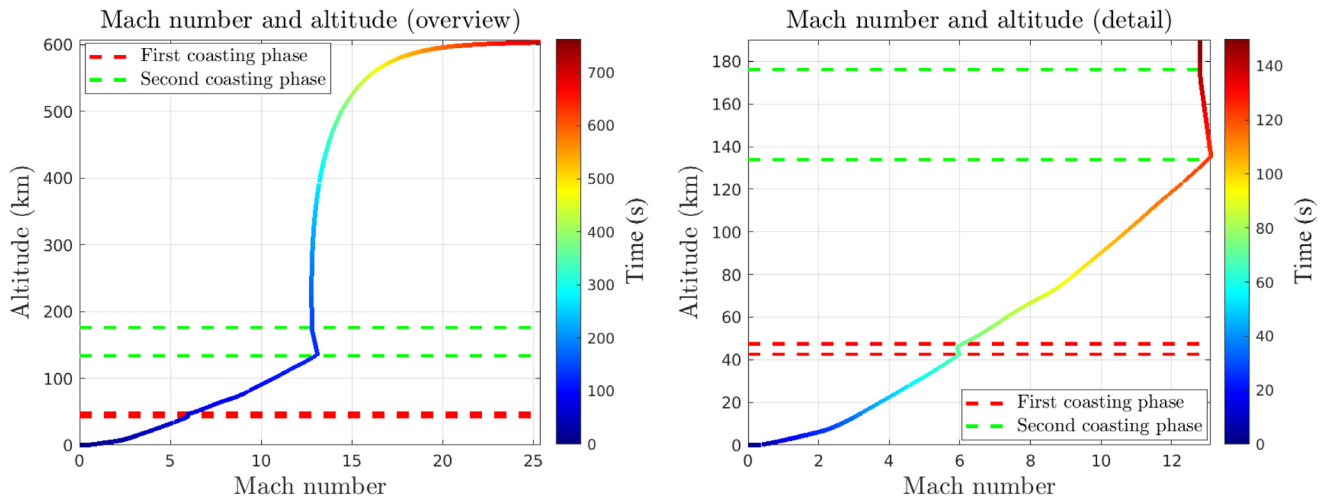
With respect to the gravity turn maneuver, the guidance scheme prioritizes load relief over drift compensation and, since dynamic pressure is unwieldy to control, the aerodynamic load will be reduced on the pitch and yaw axes by minimizing the angle of attack and the sideslip angle, respectively, for the duration of the gravity turn. Thus, deviations from the desired trajectory may accumulate towards the end of the first stage, yet this fact is alleviated during the exo-atmospheric stage.

The second phase's aim is to ensure successful orbit circularization. To do so, a version of the Unified Powered Flight Guidance (UPFG) algorithm, introduced in Brand et al. (1973), is employed to correct the drift that has accumulated at the end of the gravity turn maneuver. Note that this is achievable due to the compact configuration of the micro-launcher, which allows for sufficient trajectory corrections during the remaining 15 minutes of the mission. Once the desired orbit is achieved, with the third separation underway, cold-gas thrusters on the payload insertion module's pitch and yaw axes activate alongside those on the roll axis to bring all angles to 0.

2.2. Mission specifications

To achieve mission success, the proposed control algorithm must ensure that the design specifications enumerated in Table 1 are successfully met. The remainder of this subsection is dedicated to explaining the entries given in Table 1, to better correlate them with the mission description from Section 2.1.

In order to prevent the structural failure of the micro-launcher, the most fundamental requirement placed upon the controller is that it should render the launch vehicle's dynamics stable in closed-loop configuration, even in the face of model uncertainty, throughout the entire ascent. Classically, this translates into ensuring 6 dB gain margin and 30 deg phase margin [see Section 4.2 in Doyle et al. (1990)] to ensure robust stability. However, a more rigorous set of specifications has been chosen, namely that the closed-loop system must be robustly stable with respect to a norm-bounded class of uncertainty, while



(a) Overview representation of the entire trajectory

(b) Detailed view centered around the coasting phases

Fig. 1. Micro-launcher speed and altitude profile throughout the ascent.

Table 1
Closed-loop system requirements.

Disk gain margin	Disk phase margin	α, β settling time	$\lvert\alpha\rvert$, $\lvert\beta\rvert$ during gravity turn	θ, ψ settling time	θ, ψ overshoot	θ, ψ steady state error	$\lvert\delta_\theta\rvert$, $\lvert\delta_\psi\rvert$ during ascent	$\lvert\phi\rvert$ during ascent
≥ 3 dB	≥ 15 deg	≤ 4 s	≤ 2 deg	≤ 4 s	$\leq 5\%$	$\leq 0.1\%$	≤ 7 deg	≤ 2 deg

translating classical margins into the modern multi-loop disk margins [see Blight et al. (1994)].

Along with robust stability, another factor used to prevent loss-of-vehicle is load relief, which is intended to reduce the total loads (bending moment) by reducing the product of the dynamic pressure and the total angle of attack or sideslip angle, depending on the considered axis of motion. Both the angle of attack and the sideslip angle must be kept within ± 2 deg for the duration of the gravity turn. To account for the inclination maneuver that precedes the latter and the coasting phase which occurs during the gravity turn, a 4 second settling time after the end of both events is allowed before enforcing the constraint upon the angle of attack and the sideslip angle. Regarding the flexible-body dynamics of the vehicle, the effect of bending modes upon closed-loop behavior should be minimized.

In terms of ensuring correct orbit circularization, the various maneuvers planned by the employed guidance algorithms are interpreted as reference signals for the pitch and yaw angles. In terms of transient response properties, the closed-loop step response must have a settling time of at most 4 s, a rise time of less than 1 second, a maximum overshoot of 5% and under 0.1% steady-state tracking error. These performance requirements should be achieved by using at most ± 2.5 deg of TVC nozzle deflection from the maximum command amplitude of ± 7 deg, with an additional ± 1 deg employed for wind disturbance rejection and ± 3.5 deg kept in reserve for unplanned maneuvers and coasting correction, at the end of stage separations. The roll angle should be kept to at most ± 2 deg during the entire ascent, while limiting the amount of expended cold-gas. Finally, performance must be consistent across all possible realizations of the micro-launcher's model.

2.3. Model for pitch and yaw axes

The nonlinear and parametrically uncertain dynamical equations which fully describe the micro-launcher's 6-degree of freedom motion (Chelaru, 2018) are largely intractable for the purpose of obtaining a control system. However, controllers are a necessary component of any flight mission, as the micro-launcher's dynamics are unstable, which precludes any form of open-loop control. Thus, consider first the rigid-body dynamics of the micro-launcher on the pitch and yaw axes.

Remark 2. Due to the geometric configuration of the launch vehicle, the pitch and yaw dynamics are nominally identical when the micro-launcher's roll rate is kept small. This fact will be ensured throughout the mission by the roll control system (see Section 3.5 in the sequel) and, therefore, only the pitch axis will be referred to for the remainder of this paper.

Let the pitch-axis dynamics be given, at N instants of time along the trajectory, by the representation

$$\begin{aligned} \dot{x}(t) &= A_k x(t) + B_k u(t), \\ y(t) &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} x(t) = C x(t), \end{aligned} \tag{1}$$

where $k \in 1 : N$ indexes the instant to produce a set of realizations $(A_k, B_k, C, 0)$ [see Chapter 2 of Kailath (1980)]. For an arbitrary k, the differential part of (1) may then be expressed as

$$\begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ a_{21} & 0 & a_{23} \\ a_{31} & 0 & a_{33} \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \\ \dot{z} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} \begin{bmatrix} \delta_\theta \\ \alpha_w \end{bmatrix}. \tag{2}$$

Note that all angles are expressed in deg, all angular velocities are given in deg/s, and all translational speeds are given in m/s, while the entries of the matrices from (2) are given by

$$\begin{cases} a_{21} = H_0 \frac{C_{n\alpha}}{C_{yz}}, \\ a_{23} = -\frac{a_{21}}{V}, \\ a_{31} = \frac{F_0}{m} C_{L\alpha} + \frac{T}{m} \cos(\alpha - \delta_\theta) - \frac{V^2}{R} \sin(\theta - \alpha), \\ a_{33} = -\frac{F_0}{mV} C_{L\alpha} - \frac{T}{mV} \cos(\alpha - \delta_\theta) + \frac{g}{V} \sin(\theta - \alpha), \\ b_{21} = \frac{-T\ell}{C_{yz}} x \cos(\delta_\theta), \\ b_{22} = -a_{21}, \\ b_{31} = -\frac{T}{m} \cos(\alpha - \delta_\theta), \\ b_{32} = -\frac{F_0}{m} C_{L\alpha} - \frac{T}{m} \cos(\alpha - \delta_\theta) + g \sin(\theta - \alpha), \end{cases} \tag{3}$$

with the terms from the above relations being the various aerodynamic coefficients, geometric proportions and angular positions of the launch

vehicle, given in the nomenclature on the first page. Although essential in their own right for the purpose of modeling the micro-launcher's dynamics, the focus is placed upon the entries of the state-space matrices from (2), as these are then employed to produce the sought-after controllers.

Remark 3. The linearized equations of motion from (2), which describe the micro-launcher's rigid body dynamics, have been expressed by following the same modeling principles as in Section 1.3.2 of Gadea Díaz and Bennani (2011) [see also Du (2010) and Greensite (1970)]. The aforementioned state-space representation is obtained by first linearizing the various trigonometric terms in (9)–(10) of Chelaru (2018) and then applying a linear state-vector transformation dependent upon the micro-launcher's total velocity.

Finally, consider the thrust vectoring control (TVC) actuator that physically implements the command signal being transmitted to the first input channel of the system in (1). This is given by a stable second-order input-delayed low-pass filter

$$\mathbf{G}_{TVC}(s) = K_{TVC}\frac{\omega_{TVC}^2}{s^2 + 2\zeta_{TVC}\omega_{TVC}s + \omega_{TVC}^2}e^{-\tau_{TVC}s}, \tag{4}$$

which may readily be converted into state-space form via a first-order Padé approximation of the input delay.

Remark 4. Note that the simplified actuator dynamics [modeled as in Section 1.3.2 of Gadea Díaz and Bennani (2011)] do not explicitly describe the "Tail-Wags-Dog" effect [see, for example, (2) in Pulecchi et al. (2012)], caused by engine gimbal acceleration. This can prove particularly important for small-scale launchers and, thus, this model mismatch has been implicitly considered via the actuator's parametric uncertainties (see (11) in the sequel). Indeed, note that the variable quantities from (11) are significantly greater than the ones from (5)–(6), discussed in the sequel, in order to compensate for the unmodeled nominal dynamics. Also, the first-order Padé approximation of the input delay also serves as an adequate stand-in for the actuator's servo lag, modeled by the additional stable pole.

2.4. Uncertainty mapping

One of the main challenges in obtaining a suitable controller for the rigid-body model of the micro-launcher is given by the fact that the dynamics from (2) are parametrically uncertain, which significantly complicates the issue of closed-loop internal stability [for the exact definition, see Section 5.1 of Vidyasagar (1985)].

Of the physical parameters in (3), which influence the launch vehicle's dynamics, only 6 are considered to have uncertain values during the process of linearizing the micro-launcher's model. At any instant k, the uncertain values of these parameters (denoted by the superscript $(\cdot)^{\Delta}$) are derived from a two-sided normal distribution at the 3 standard deviation level [see Appendix D3 in Garafend (2006)], so that they represent 99.73% of the expected population, respectively,

$$\begin{aligned} C_{L\alpha,k}^{\Delta} &= C_{L\alpha,k}\left(1 \pm 3\sigma_A\right), & C_{n\alpha,k}^{\Delta} &= C_{n\alpha,k}\left(1 \pm 3\sigma_A\right), \\ C_{yz,k}^{\Delta} &= C_{yz,k}\left(1 \pm 3\sigma_C\right), & m_k^{\Delta} &= m_k\left(1 \pm 3\sigma_m\right), \\ \ell_k^{\Delta} &= \ell_k\left(1 \pm 3\sigma_\ell\right), & T_k^{\Delta} &= T_k\left(1 \pm 3\sigma_T\right), \end{aligned} \tag{5}$$

where the standard deviations for aerodynamic coefficients (σ_A), moment of inertia (σ_C), mass (σ_m), distance (σ_ℓ), and thrust force (σ_T) of the micro-launcher's rigid-body are

$$\begin{gathered} \sigma_A = \sqrt{3}\%, \quad \sigma_C = \sqrt{3}\%, \quad \sigma_m = \sqrt{2}\%, \\ \sigma_\ell = 1\%, \quad \sigma_T = 1\%. \end{gathered} \tag{6}$$

Remark 5. Although the ranges defined in (5) and (6) are not immediately indicative of significant model uncertainty, note however the nonlinearly-coupled fashion in which these uncertain parameters combine among themselves in (3). By grouping up these couplings, which are formed from several parameters, a simplified modeling process [see Section 10.3 of Zhou et al. (1996)] was carried out, for the purpose of employing μ-synthesis [see Chapter 11 of Zhou et al. (1996)]. Yet, this representation of uncertainty has proven to be excessively conservative, while the nonlinear coupling between the 10 sources of uncertainty, from (5) and (11) in the sequel, significantly increased the number of linear uncertain parameters and also the dimension of the generalized system. Due to numerical issues exhibited by the latter when employing the D-K iteration [see Section 11.4 of Zhou et al. (1996)], this representation of uncertainty has proven to be intractable for the considered application.

In order to obtain a tractable representation for the system's uncertainty, a thousand random samples of the uncertain parameters from (5) were generated and a set of possible realizations for the micro-launcher's pitch dynamics was constructed via (2) and (3). Refer to Fig. 2, which showcases, in its leftmost two columns, the frequency response of the obtained set of systems at $k = 1$, which corresponds to $t = 1.4$ s after lift-off.

Remark 6. The frequency diagrams showcased in Fig. 2 are illustrative of the fact that uncertainty naturally separates along the two input channels of the rigid-body pitch model. This allows for a particularly useful representation for mapping uncertainty, namely in an input-multiplicative form, which not only enables less conservative bounds on uncertainty than, for example, an output-multiplicative form, but also allows us to safely bypass the wind incidence channel uncertainty when considering closed-loop stability (see Remark 11 in the sequel), alleviating the trade-off between stability and performance.

Let

$$\mathbf{G}_k(s) := C(sI - A_k)^{-1}B_k = \left[\begin{array}{c|c} \mathbf{G}_{11}^k & \mathbf{G}_{12}^k \\ \hline \mathbf{G}_{21}^k & \mathbf{G}_{22}^k \end{array}\right] \tag{7}$$

denote the nominal transfer function matrix [abbreviated TFM, see chapter 6 of Kailath (1980)] of system (1). Denote the set of sampled realizations at instant k by $\mathcal{U}\left(\mathbf{G}_k\right)$ and let the associated class of uncertain systems (indicated by the superscript $(\cdot)^{\Delta}$) at instant k be given by the input-multiplicative representation

$$\mathbf{G}_k^{\Delta}(s) := \mathbf{G}_k(s)\begin{bmatrix} 1 + \mathbf{\Delta}_1(s)\mathbf{W}_\delta^k(s) & 0 \\ 0 & 1 + \mathbf{\Delta}_2(s)\mathbf{W}_{\alpha_w}^k(s) \end{bmatrix}, \tag{8}$$

where $\mathbf{\Delta}_1, \mathbf{\Delta}_2$ are elements in $\mathcal{RH}_\infty^{1\times1}(s)$ (the set of all stable transfer functions) with the $\mathcal{H}_\infty$ norm [see Bruinsma and Steinbuch (1990) for computational details] defined as $\|\mathbf{\Delta}_i\|_\infty := \sup_{\omega\in\mathbb{R}\cup\{\infty\}}\overline{\sigma}\left(\mathbf{\Delta}_i(j\omega)\right) \leq 1, i \in 1:2$; $\overline{\sigma}(\cdot)$ denotes the maximum singular value of a matrix and $\underline{\sigma}(\cdot)$ is the minimum one [see section 2.8 of Zhou et al. (1996)].

Remark 7. The chief reasoning behind this process of modeling uncertainty is that, by analyzing the realizations from $\mathcal{U}\left(\mathbf{G}_k\right)$, one may compute frequency-dependent bounds upon the variations in gain and bandwidth which are induced by the previously discussed sources of uncertainty. Using these bounds, it is possible to formulate criteria maintaining tracking and actuation performance, even in the face of uncertainty, while also ensuring closed-loop stability with the guarantees provided by a set of disk margins, to be discussed in the sequel.

The connection between $\mathcal{U}\left(\mathbf{G}_k\right)$ and $\mathbf{G}_k^{\Delta}(s)$ is made through the stable filters $\mathbf{W}_\delta^k(s)$ and $\mathbf{W}_{\alpha_w}^k(s)$, that are illustrated in the rightmost column of Fig. 2 and which constitute the solution to the optimal filter-fitting problems

$$\begin{cases} \inf_{\mathbf{W}_\delta^k\in\mathcal{RH}_\infty^{1\times1}(s)} \int_{-\infty}^{+\infty} \left(\left|\mathbf{W}_\delta^k(j\omega)\right| - \mathcal{G}_k^1(\omega)\right) \mathrm{d}\omega, \\ \text{subject to} \quad \left|\mathbf{W}_\delta^k(j\omega)\right| \geq \mathcal{G}_k^1(\omega), \forall\omega \in \mathbb{R}\cup\{\infty\}, \end{cases} \\ \begin{cases} \inf_{\mathbf{W}_{\alpha_w}^k\in\mathcal{RH}_\infty^{1\times1}(s)} \int_{-\infty}^{+\infty} \left(\left|\mathbf{W}_{\alpha_w}^k(j\omega)\right| - \mathcal{G}_k^2(\omega)\right) \mathrm{d}\omega, \\ \text{subject to} \quad \left|\mathbf{W}_{\alpha_w}^k(j\omega)\right| \geq \mathcal{G}_k^2(\omega), \forall\omega \in \mathbb{R}\cup\{\infty\}, \end{cases} \tag{9}$$

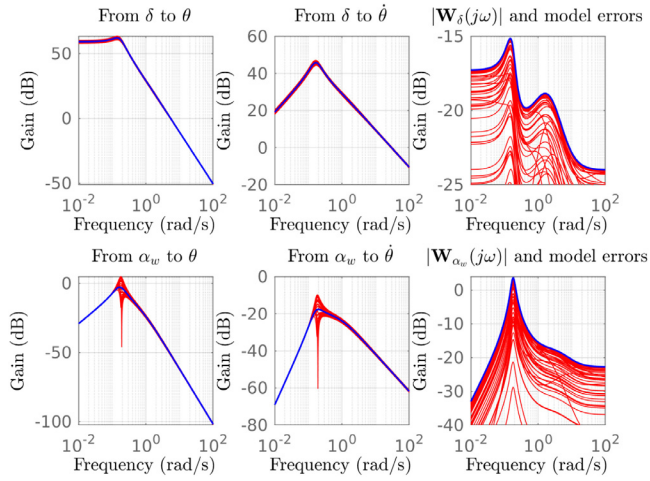


Fig. 2. Uncertain plant and associated weights at 1.4 s after lift-off. Uncertain systems are displayed in red, while nominal ones are shown in blue. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

where $\mathcal{G}_k^i(\omega)$, with $i \in 1:2$, are given at every ω by

$$\mathcal{G}_k^i(\omega) := \max\left\{ \max_{\hat{\mathbf{G}}_k \in \mathcal{U}(\mathbf{G}_k)} \left| \frac{\hat{\mathbf{G}}_{1i}^k(j\omega)}{\mathbf{G}_{1i}^k(j\omega)} - 1 \right|, \max_{\hat{\mathbf{G}}_k \in \mathcal{U}(\mathbf{G}_k)} \left| \frac{\hat{\mathbf{G}}_{2i}^k(j\omega)}{\mathbf{G}_{2i}^k(j\omega)} - 1 \right| \right\}, \quad (10)$$

where the terms $\left| \frac{\hat{\mathbf{G}}_{1i}^k(j\omega)}{\mathbf{G}_{1i}^k(j\omega)} - 1 \right|$ and $\left| \frac{\hat{\mathbf{G}}_{2i}^k(j\omega)}{\mathbf{G}_{2i}^k(j\omega)} - 1 \right|$ represent the model errors which are shown in the rightmost column of Fig. 2 as being upper bounded by the computed uncertainty weights. Each of the optimization problems given by (9) aims to find a stable filter (of imposed order) whose magnitude represents the least upper bound for the model errors given in (10).

Finally, the same procedure is applied to the TVC actuator model from (4), where the four uncertain parameters are also considered normally distributed and are given below with the ± 3 standard deviations (expressed in absolute values), respectively

$$\begin{aligned} K_{TVC}^{\Delta} &= 1 \pm 0.2, & \tau_{TVC}^{\Delta} &= 0.015 \pm 0.003, \\ \omega_{TVC}^{\Delta} &= 120 \pm 24, & \zeta_{TVC}^{\Delta} &= 0.7 \pm 0.07, \end{aligned} \quad (11)$$

to obtain

$$\mathbf{G}_{TVC}^{\Delta}(s) = \mathbf{G}_{TVC}(s)(1 + \mathbf{\Delta}_3(s)\mathbf{W}_{TVC}(s)), \quad (12)$$

where $\mathbf{\Delta}_3 \in \mathcal{RH}_\infty^{1\times 1}(s)$ and $\|\mathbf{\Delta}_3\|_\infty \le 1$, while $\mathbf{W}_{TVC}(s)$ is the deduced stable filter for actuator uncertainty.

2.5. Bending modes

Aside from the rigid-body dynamics introduced in Section 2.3 and expanded to account for parametric uncertainty in Section 2.4, the pitch and yaw motions are also affected by the micro-launcher's flexible-body dynamics, which manifest via the so-called bending modes. These can be modeled, on the pitch axis, for example, as a set of N_b signals that act additively upon the rigid-body pitch angle (θ) and which are approximated [see Noll et al. (1970) for a comprehensive treatment of the subject] by second-order ordinary differential equations

$$\ddot{\theta}_i + 2\zeta_i\omega_i\dot{\theta}_i + \omega_i^2\theta_i = \frac{180}{\pi}K_\theta \sin(\delta_\theta), \; i \in 1:N_b.$$

Each bending mode (θ_i), expressed in degrees, depends upon the TVC deflection angle (δ_θ) and a bending constant (K_θ), which is a function of mass, thrust force and of the distance between the mass and the pressure centers. Note that the damping factors of these modes are usually quite small, between 0.005 and 0.02, while the natural bending frequencies of the micro-launcher (ω_i) depend upon the latter's geometric configuration.

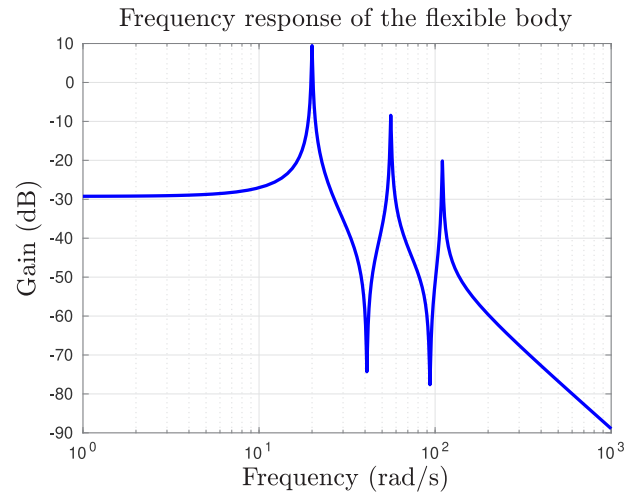


Fig. 3. The three bending modes at 20, 56 and 110 rad/s for $t = 7$ s.

To showcase this, consider the launch vehicle's first $N_b = 3$ bending modes at $t = 7$ s, where $K_\theta \approx 1.7$ and (for the simulations given in Section 5) $\delta_\theta = 1$ deg. The frequency response of the launcher's flexible body, obtained by linearizing $\sin(\delta_\theta)$ as δ_θ due to the TVC command limitations ($\delta_\theta \ll 7$ deg at all times, see Fig. 9(b) in the sequel), is shown in Fig. 3.

2.6. Nonlinear roll model

The primary aim of the control system will be to regulate the pitch and yaw angles as indicated by the guidance algorithm, in order to ensure mission success. However, lack of adequate control on the roll-axis channel can lead to mission failure due to the undesirable phenomenon of rotational momentum cross-coupling in the micro-launcher's attitude [see Chapter 4 of Du (2010)]. Let the roll dynamics be controlled via a set of on–off-reverse cold-gas thrusters and consider the nonlinear dynamics

$$\begin{aligned} \ddot{\phi} &= \frac{T_{THR}}{C_x}\delta_{THR}, & \dot{\delta}_{THR} &= \frac{u_{THR}^{\phi} - \delta_{THR}}{\tau_{THR}}, \\ \dot{\theta}_{tot} &= \frac{C_x - C_{yz}}{C_{yz}}\psi_{tot}\phi + \dot{\theta}, & \dot{\psi}_{tot} &= \frac{C_{yz} - C_x}{C_{yz}}\theta_{tot}\phi + \dot{\psi}. \end{aligned} \quad (13)$$

Note also that the cold-gas thruster's time-delayed relay dynamics are reasonably approximated by a first-order linear system.

3. Control system synthesis

3.1. Control system architecture

To satisfy all mission requirements presented in Section 2.2, the main control problem has been broken down into multiple sub-problems (see Fig. 4 on the next page), with each sub-solution addressing a particular aspect of the mission.

At the very foundation of the control subroutine are the TVC controllers for the pitch and yaw axes. It is their purpose to take the angular values and velocities $(\theta, \dot{\theta}, \psi, \dot{\psi})$ for their respective channels from the Inertial Measurement Unit (IMU) and compute appropriate deflection angles $(\delta_\theta, \delta_\psi)$ for the TVC nozzle, to guarantee that the pitch and yaw angles track their prescribed references (θ_r, ψ_r), computed by the guidance algorithm.

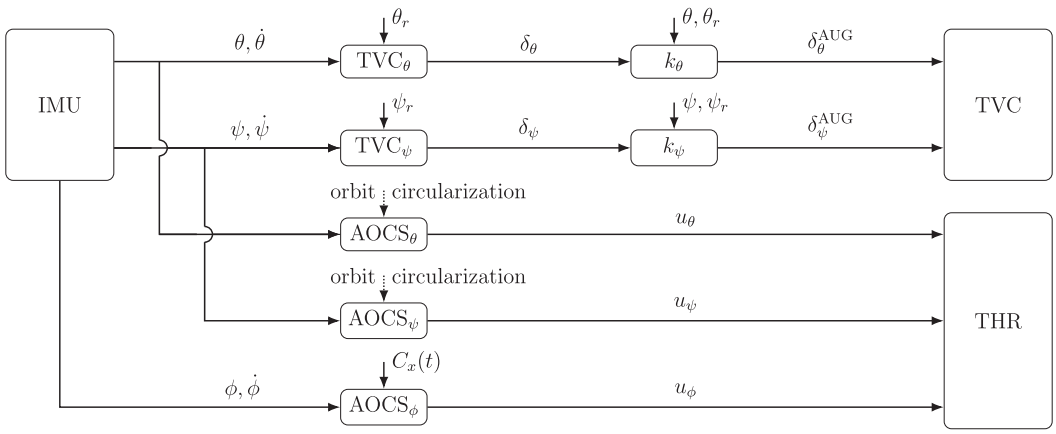


Fig. 4. Control system architecture diagram.

Remark 8. Note that only attitude feedback is employed for command computation, without using any drift rate information. Given that the latter's measurements from the IMU are influenced by both the launch vehicle's angular acceleration and (the uncertain) distance between the IMU and the micro-launcher's center of mass [see (16) in Stoica et al. (2019)], the proposed feedback scheme is both simpler and sufficient to ensure mission success (see Section 5 in the sequel). Moreover, by exploiting the fact that the pitch and yaw dynamics are nominally identical (recall Remark 2) and subject to the same class of uncertainty, one can utilize the same TFM for both controllers on the two axes ($\mathrm{TVC}_\theta \equiv \mathrm{TVC}_\psi$). A final aspect that concerns the TVC controllers is the type of gain-scheduling employed for the ascent mission. Since the micro-launcher's nominal model does not vary significantly on the pitch and yaw axes between stage separations, it is possible to employ only one such controller during each of the three ascent stages. The commutation occurs during the two coasting phases, thus eliminating the need to interpolate between multiple controllers.

Consider now the bending modes that make up the flexible-body model on the pitch and yaw axes. Although bending mode compensation is also regularly solved via frequency-based optimization (Ganet-Schoeller et al., 2017), another type of procedure has been chosen for the proposed solution. Let the command signals computed by the latter ones ($\delta_\theta, \delta_\psi$) be weighted by adaptive augmentation gains (k_θ, k_ψ) before the new commands ($\delta_\theta^{\mathrm{AUG}}, \delta_\psi^{\mathrm{AUG}}$) are transmitted to the TVC actuators. It is the purpose of these adaptive gains to analyze commands and IMU readings for bending-related phenomena and decrease command amplitudes.

Remark 9. Since the TVC controllers are not explicitly designed to have any bending filter capabilities, the interaction between the rigid-body and flexible-body dynamics may cause instability in the absence of an augmentation scheme. However, the latter is always active throughout the ascent, no matter what flight scenario the micro-launcher is in, and that its main purpose (see also Section 6.2 in the sequel) is to adjust the behavior of the TVC controllers whenever the aforementioned interactions begin to compromise closed-loop stability.

There remains the issue of minimizing roll-induced disturbance and controlling the payload insertion module's attitude once orbit circularization is achieved and TVC actuation becomes unavailable, following the third separation stage. These tasks are handled by an Attitude and Orbit Control System (AOCS), shown in Fig. 4 as operating in parallel with the TVC controllers. While the roll-channel AOCS (AOCS_ϕ) does indeed receive feedback ($\phi, \dot{\phi}$) from the IMU and triggers the cold-gas thrusters during the entire ascent, the pitch and yaw AOCS subroutines ($\mathrm{AOCS}_\theta, \mathrm{AOCS}_\psi$) are brought online only after orbit circularization is achieved. Note that all the command signals generated by the AOCS subsystem (u_θ, u_ψ, u_ϕ) are computed via sliding-mode control [see Section 10.1 in Khalil (2015)] based upon nonlinear dynamics of type (13) and converted to on–off-reverse values through pulse-width-pulse-frequency (PWPF) modulation [see Song et al. (1999)].

Remark 10. The pitch and yaw AOCS do not require gain-scheduling, as they are tuned for the payload insertion module's dynamics, while the roll-channel AOCS is dependent upon the value of the roll moment of inertia, $C_x(t)$, to account for variations due to stage separation and fuel consumption.

The remaining part of Section 3 is dedicated to detailing the design procedures used to obtain each component in the presented architecture. Section 3.2 presents the mixed-sensitivity approach employed for the first and second stage TVC controller. Section 3.3 concerns the singular value loopshaping technique that produces the third stage TVC controller. Section 3.4 elaborates upon the composition of the adaptive gain scheme and the role each component plays in suppressing bending-related phenomena. Finally, Section 3.5 details the implementation of the roll control system and of the payload insertion command algorithm.

3.2. First and second stage synthesis procedure

During the atmospheric phase of the ascent, which is covered by the micro-launcher's first and second stages, the launch vehicle's model is employed for mixed-sensitivity synthesis (see Fig. 5). A simpler version of this procedure has been validated in Diaconescu et al. (2019) for only the first 17 s of the first stage (until gravity turn), albeit without the robust performance guarantees which will be discussed in Section 4.

In this approach, frequency-based weights are assigned to the tracking error signal (e), the command signal (δ_θ) and the various signals related to the sources of uncertainty (w_1, w_3, z_1, z_3), and $\mathcal{H}_\infty$ optimization is performed to obtain both robust stability, with respect to the class of systems given by (8) and (12), along with the desired transient response properties. Let $\mathbf{T}_{yu}$ denote a generic transfer function from a signal u to a signal y. Then, the mixed-sensitivity procedure is deemed to be successful if the following conditions are met:

(C_1) All transfer functions in Fig. 5 from exogenous signals to those internal to the control loop are stable;
(C_2) $\left\|\mathbf{T}_{z_1 w_1}\right\|_\infty < 1$;
(C_3) $\left\|\mathbf{T}_{z_3 w_3}\right\|_\infty < 1$;
(C_4) $\left\|\mathbf{T}_{e_w \theta_r}\right\|_\infty < 1$, for a given stable frequency weight $\mathbf{W}_e$ that quantifies tracking performance;
(C_5) $\left\|\mathbf{T}_{\delta_w \theta_r}\right\|_\infty < 1$, for a given stable frequency weight $\mathbf{W}_{cmd}$ that quantifies command limitation.

$$\begin{bmatrix} z_1 \\ z_3 \\ e_w \\ \delta_w \\ \hline e \\ \dot{\theta} \end{bmatrix} = \underbrace{\left[\begin{array}{cccc|c} 0 & \mathbf{W}_\delta \mathbf{G}_{TVC} & 0 & 0 & \mathbf{W}_\delta \mathbf{G}_{TVC} \\ 0 & 0 & 0 & 0 & \mathbf{W}_{TVC} \\ -\mathbf{W}_e \mathbf{G}_{11}^{nom} & -\mathbf{W}_e \mathbf{G}_{11}^{nom} \mathbf{G}_{TVC} & \mathbf{W}_e & -\mathbf{W}_e \mathbf{G}_{12}^{nom} & -\mathbf{W}_e \mathbf{G}_{11}^{nom} \mathbf{G}_{TVC} \\ 0 & 0 & 0 & 0 & \mathbf{W}_{cmd} \\ \hline -\mathbf{G}_{11}^{nom} & -\mathbf{G}_{11}^{nom} \mathbf{G}_{TVC} & I & -\mathbf{G}_{12}^{nom} & -\mathbf{G}_{11}^{nom} \mathbf{G}_{TVC} \\ \mathbf{G}_{21}^{nom} & \mathbf{G}_{21}^{nom} \mathbf{G}_{TVC} & 0 & \mathbf{G}_{22}^{nom} & \mathbf{G}_{21}^{nom} \mathbf{G}_{TVC} \end{array}\right]}_{\mathbf{G}_{gen}} \begin{bmatrix} w_1 \\ w_3 \\ \theta_r \\ \alpha_w \\ \hline \delta_\theta \end{bmatrix} \tag{14}$$

Box I.

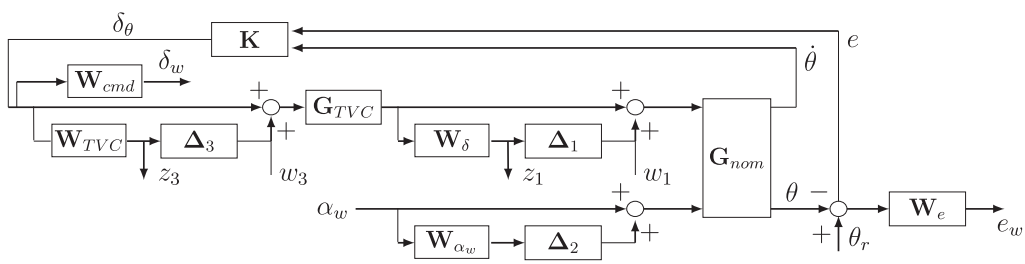


Fig. 5. Mixed-sensitivity weighting scheme.

Remark 11. Condition (C_1) is equivalent to closed-loop internal stability which, combined with conditions $(C_2)-(C_3)$ and the decoupled nature of the uncertainty, guarantees via the Small Gain Theorem [see section 9.2 in Zhou et al. (1996)] that the closed-loop system is robustly stable. Note that, due to the configuration in Fig. 5, the wind incidence channel uncertainty (as quantified by $\mathbf{W}_{\alpha_w}$) has no impact on closed-loop stability and only modulates the frequency response of the wind disturbance. By ensuring that the bandwidths of $\mathbf{W}_e$ and $\mathbf{W}_{cmd}$ include that of the wind's frequency profile, satisfying conditions $(C_4)-(C_5)$ (which signify nominal tracking performance and nominal command limitation) will also ensure atmospheric disturbance rejection, via the shaping of the sensitivity function.

In order to simultaneously satisfy conditions $(C_1)-(C_5)$, define the $\mathcal{H}_\infty$ optimization problem given by the generalized system $\mathbf{G}_{gen}(s)$ from (14) (as given in Box I), with the column vectors containing the Laplace transforms of signals appearing in Fig. 5. The weights $\mathbf{W}_\delta$ and $\mathbf{W}_{TVC}$ have been deduced as in Section 2.4 and $\mathbf{G}_{nom}(s) = \mathbf{G}_k(s)$ is the nominal model at instant k employed for synthesis, with $\mathbf{G}_{gen}(s)$ partitioned conformally with (14), as follows

$$\mathbf{G}_{nom}(s) = \left[\begin{array}{c|c} \mathbf{G}_{11}^{nom} & \mathbf{G}_{12}^{nom} \\ \hline \mathbf{G}_{21}^{nom} & \mathbf{G}_{22}^{nom} \end{array}\right], \quad \mathbf{G}_{gen}(s) = \left[\begin{array}{c|c} \mathbf{G}_{11}^{gen} & \mathbf{G}_{12}^{gen} \\ \hline \mathbf{G}_{21}^{gen} & \mathbf{G}_{22}^{gen} \end{array}\right].$$

Then, to ensure $(C_1)-(C_5)$, it is sufficient to find a controller $\mathbf{K}(s)$ which makes the lower linear fractional transformation (LLFT)

$$\mathcal{F}_\ell(\mathbf{G}_{gen}, \mathbf{K}) = \mathbf{G}_{11}^{gen} + \mathbf{G}_{12}^{gen}\mathbf{K}(I - \mathbf{G}_{22}^{gen}\mathbf{K})^{-1}\mathbf{G}_{21}^{gen}$$

internally stable and guarantees that $\left\|\mathcal{F}_\ell(\mathbf{G}_{gen}, \mathbf{K})\right\|_\infty < 1$. Note that, regardless of the chosen tuning weights, $\mathbf{W}_e$ and $\mathbf{W}_{cmd}$, the final row of $\mathbf{G}_{gen}(s)$ (corresponding to the $\dot{\theta}$ output feedback) is composed only of strictly proper or null entries and, thus, the feedthrough matrix of $\mathbf{G}_{21}^{gen}(s)$ does not have full row rank. This negates the standard regularity assumptions required for the dual Riccati equation solution for norm optimization and makes the classical approach [see Doyle et al. (1989)] infeasible. Thus, the LMI-based approach [see Gahinet and Apkarian (1994)] has been employed to solve this singular problem.

Remark 12. Satisfactory results for both the first and the second micro-launcher stages have been achieved by employing

$$\mathbf{W}_e(s) = \frac{0.005s + 1}{s + 0.001}, \quad \mathbf{W}_{cmd}(s) = \frac{s}{s + 100}.$$

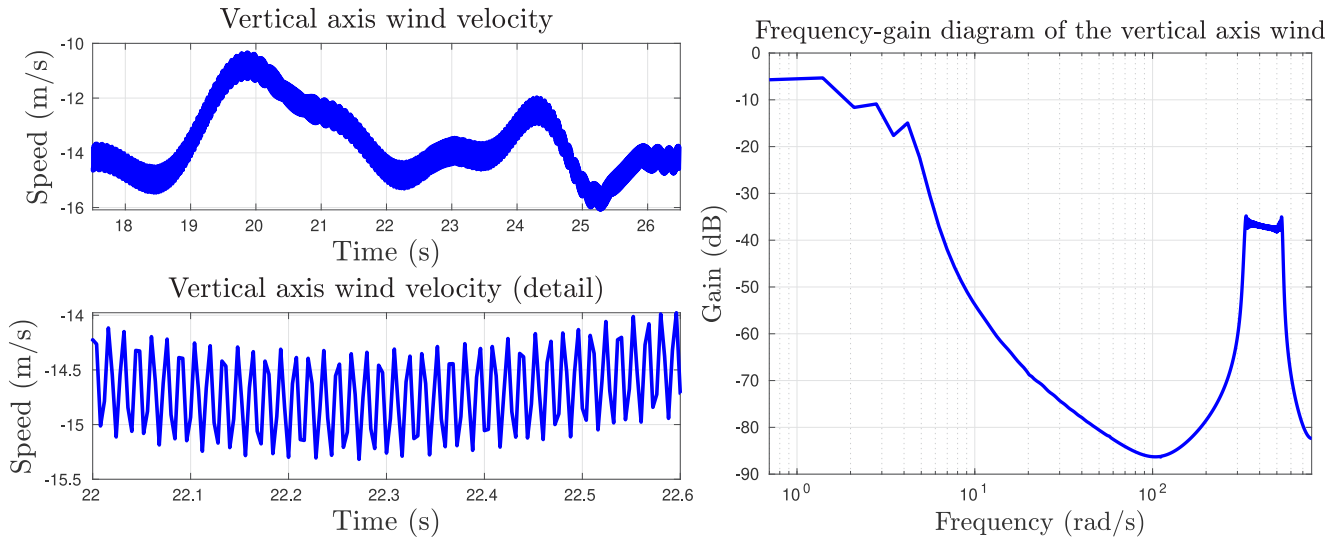
For good steady-state tracking, $\mathbf{W}_e$ must possess a large low-frequency gain, while a large enough bandwidth guarantees rapid attenuation of tracking error and good wind disturbance rejection. Refer to Fig. 6(a), where vertical axis windspeed (affecting the pitch angle) has been generated using Dryden filters, with moderate turbulence. Using a sampling rate of 250 Hz and applying a Fast Fourier Transform algorithm, one obtains the positive frequency part of the double–sided amplitude spectrum, shown in Fig. 6(b). By analyzing the latter, notice that the wind signal has significant harmonics only up to around 1 rad/s. This has been chosen as the bandwidth of $\mathbf{W}_e$, in order to penalize the sensitivity function and ensure satisfactory rejection of the main wind harmonics. It is also desirable to avoid imposing any constraints upon steady-state actuation, and therefore $\mathbf{W}_{cmd}$ is chosen as a high-pass filter with a zero in $s = 0$, with the added benefit of penalizing high-frequency actuation.

Remark 13. Note that $\mathbf{W}_e$ has a very low magnitude at high frequency which, in combination with $\mathbf{W}_{cmd}$ having high magnitude in that frequency range, places a valid constraint only upon the controller's high-frequency gain, while allowing the sensitivity function to (asymptotically) approach unit gain. The resulting controllers naturally have a low-pass frequency characteristic, which aids in filtering out high-frequency measurement noise from the IMU. By limiting the bandwidth of the tuning weights, some performance is sacrificed in order to prevent any significant "waterbed" effects, which cause oscillations at high frequencies [see Section 6.2 of Doyle et al. (1990)].

3.3. Third stage synthesis procedure

Although the mixed-sensitivity approach produces satisfactory results for the first two stages, the micro-launcher's dynamical model during the third stage prevents the synthesis procedure from being employed. The reason for this is that the third stage evolves in exo-atmospheric conditions and, due to the lack of air friction, the open right-half poles of the dynamical model on the pitch and yaw axes migrate into the origin of the complex plane. Indeed, due to the conservation of impulse and momentum, the rigid-body model becomes a double integrator.

To counteract the poor numerical conditioning of formulating the control problem via the mixed-sensitivity approach, the design procedure switches to a singular value loopshaping procedure [see McFarlane



(a) Time-domain representation of windspeed

(b) Frequency-domain representation of windspeed

Fig. 6. Time–frequency analysis of turbulence-affected vertical windspeed.

and Glover (1992), Papageorgiou and Glover (1999) and also Voinot et al. (2003) for a contrasting approach]. This procedure is a direct application of the so-called robust stabilization problem [see Glover and McFarlane (1989)] and consists in finding two stable frequency weights, $\mathbf{W}_1$ and $\mathbf{W}_2$, such that it is possible to define a "shaped" plant

$$\mathbf{G}_s = \mathbf{W}_2\mathbf{G}_{nom}\mathbf{G}_{TVC}\mathbf{W}_1, \tag{15}$$

whose singular value frequency profile is used to represent the desired shape of the open-loop system's frequency response. For a stabilizing controller $\mathbf{K}_s$, the stability margin of $\mathbf{G}_s$ in closed-loop configuration with $\mathbf{K}_s$ is

$$b(\mathbf{G}_s, \mathbf{K}_s) = \left\| \begin{bmatrix} I \\ \mathbf{K}_s \end{bmatrix} (I - \mathbf{G}_s\mathbf{K}_s)^{-1} \begin{bmatrix} I & \mathbf{G}_s \end{bmatrix} \right\|_\infty^{-1}. \tag{16}$$

Then, for some feasible $\gamma > 1$, define first

$$\mathbf{K}_3 = \mathbf{W}_1\mathbf{K}_s\mathbf{W}_2, \quad \mathbf{G}_{tot} = \mathbf{G}_{nom}\mathbf{G}_{TVC}. \tag{17}$$

By Corollaries 3.2 and 3.4 in McFarlane and Glover (1992), ensuring $b(\mathbf{G}_s, \mathbf{K}_s)^{-1} < \gamma$ also enables one to write that

$$\underline{\sigma}(\mathbf{G}_{tot}\mathbf{K}_3)(j\omega) \gtrapprox \frac{1}{\gamma}\underline{\sigma}(\mathbf{G}_s)(j\omega), \text{ for } \omega \to 0 \text{ and } \frac{\overline{\sigma}(\mathbf{W}_2)}{\underline{\sigma}(\mathbf{W}_2)}(0) \approx 1,$$

$$\overline{\sigma}(\mathbf{G}_{tot}\mathbf{K}_3)(j\omega) \lessapprox \gamma\overline{\sigma}(\mathbf{G}_s)(j\omega), \text{ for } \omega \to \infty \text{ and } \frac{\overline{\sigma}(\mathbf{W}_1)}{\underline{\sigma}(\mathbf{W}_1)}(\infty) \approx 1,$$

due to the loopshape imparted by the double integrator contained in $\mathbf{G}_{nom}$. Since the problem of finding $\mathbf{K}_s$ and (by extension) $\mathbf{K}_3$ is always nonsingular, then it is easily solvable via the dual Riccati equation approach. However, the procedure for finding the two tuning weights is considerably more time-consuming than the mixed-sensitivity approach.

Remark 14. The singular value loopshaping approach offers no explicit means of ensuring closed-loop robust stability. To account for this, the uncertainty must first shift from an input-multiplicative formulation to an output-multiplicative one. Recall (8), (12) and (17) to define

$$\mathbf{G}_{tot}^{\Delta} = (1 + \mathbf{W}_{out}\mathbf{\Delta})I_2\mathbf{G}_{tot},$$

where $\mathbf{\Delta} \in \mathcal{RH}_\infty^{1\times1}(s)$ and $\|\mathbf{\Delta}\|_\infty \leq 1$, along with $|\mathbf{W}_{out}(j\omega)| = \max\{|\mathbf{W}_\delta(j\omega)|, |\mathbf{W}_{TVC}(j\omega)|\}$, $\forall\omega \in \mathbb{R} \cup \{\infty\}$. Then, an approximate condition for closed-loop robust stability is

$$\overline{\gamma}_{pred}\overline{\sigma}(\mathbf{G}_s)(j\omega) < \frac{1}{\overline{\sigma}\left(\mathbf{W}_{out}\right)(j\omega)}, \forall\omega, \text{ where } \overline{\sigma}(\mathbf{W}_{out})(j\omega) \gg 1,$$

and $\overline{\gamma}_{pred}$ is an upper bound for the predicted value of γ. Given the low amount of system uncertainty during the third stage, the above condition is proven, in simulation, to be sufficient. Even so, the robust stability cost will be checked in a rigorous manner post-synthesis, as detailed in Section 4.1.

3.4. Adaptive gain augmentation

The adaptive gain factor, $k_\theta > 0$, which acts upon the TVC controller output (δ_θ) in Fig. 4 is the key instrument used in limiting the effects of the micro-launcher bending modes and is represented by a time-dependent function that is lower bounded by the value of $k_\theta^{min} > 0$. Moreover, k_θ is obtained via a simplified form of the procedure introduced in Orr et al. (2014), Orr and Zwieten (2012) and it is governed by the system of differential–algebraic equations

$$\begin{aligned} k_\theta &= k_\theta^{min} + k_a, \qquad \dot{k}_a = \underbrace{f_e}_{\text{error factor}} + \underbrace{f_s}_{\text{spectral factor}} + \underbrace{f_\ell}_{\text{leakage factor}}, \\ f_e &= \tfrac{k_a^{max} - k_a}{k_a^{max}} a e_r^2, \qquad f_s = -\alpha k_a y_s, \qquad f_\ell = -\beta(k_\theta - 1). \end{aligned} \tag{18}$$

The newly introduced terms [see also Trotta et al. (2020), Angelov et al. (2018), Navarro-Tapia et al. (2019), Pang et al. (2021) regarding various tuning strategies for them] are now defined, in turn. The error factor, f_e, is tasked with increasing the adaptive gain when performance starts suffering as a result of limiting actuation. The weighted error signal for the pitch axis is given by

$$e_r = \dot{\gamma}_\theta + c\gamma_\theta, \; c > 0 \text{ and } \gamma_\theta = \theta - \theta_r,$$

where γ_θ is the tracking error and the constant c is used to tune sensitivity to performance degradation, while also filtering out any impulsive behaviors in γ_θ. The error gain, $a > 0$ from (18), is a tuning constant and the logistic factor $\frac{k_a^{max} - k_a}{k_a^{max}}$ is meant to ensure an upper bound of $k_\theta^{max} = k_\theta^{min} + k_a^{max}$.

The spectral factor, f_s, is responsible for decreasing the adaptive gain when the command signals begin to show oscillations indicative of uncompensated micro-launcher bending. This is done via the signal y_s from (18) which is weighted by k_a along with the spectral tuning gain, $\alpha > 0$, and is defined, taking Laplace transforms of all signals, as

$$y_s = \mathbf{G}_{lp}y_{hp}^2 \quad \text{and} \quad y_{hp} = \mathbf{G}_{hp}\delta_\theta,$$

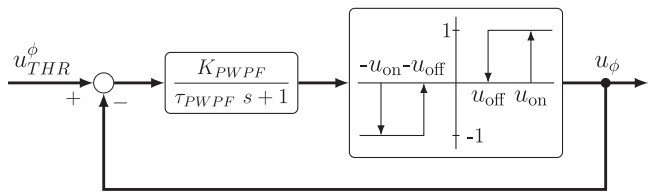


Fig. 7. Pulse-width pulse-frequency (PWPF) modulator.

where δ_θ is the command signal computed by the pitch-axis controller (see Fig. 4); $\mathbf{G}_{lp}$ along with $\mathbf{G}_{hp}$ are a low-pass and, respectively, a high-pass filter, meant to isolate the harmonics of the command signal that cannot be attributed to any rigid-body dynamics. As such, the bandwidth of $\mathbf{G}_{lp}$ must match that of the closed-loop system and the bandwidth of $\mathbf{G}_{hp}$ should be at least twice as wide.

Finally, the leakage factor, f_ℓ, is meant to reset the value of k_θ back to 1, thus restoring nominal actuation, whenever f_s indicates that no bending occurs. The leakage gain, $\beta > 0$, controls how quickly the gain factor returns to normal.

Remark 15. Since $k_\theta^{max} = 1$ is required for nominal actuation, it follows naturally that $k_a^{max} = 1 - k_\theta^{min}$. However, the decrease in gain at the plant's input may compromise closed-loop stability, since the gain was not accounted for while designing the TVC controllers. This is solved by guaranteeing a set of disk margins, which generalize classical margins for multiple-input multiple-output systems. For reasons that will be discussed in Section 4.1, it is imperative to choose $k_\theta^{min} = \frac{1}{\sqrt{2}}$.

3.5. Attitude and orbit control system

In order to prevent pitch and yaw disturbance caused by cross-coupling between these axes and motion on the roll axis during the micro-launcher's ascent, the launch vehicle's cold-gas thrusters must be employed to minimize the latter's roll angle. Recall first the roll-channel dynamics from (13), and then consider the following sliding function

$$\sigma(t) = \ddot{\phi}(t) + c_1\dot{\phi}(t) + c_0\phi(t), \tag{19}$$

and take $c_0, c_1 > 0$ to ensure that the above differential polynomial is Hurwitz [see also Chang et al. (2002), Orosco et al. (2018), Peng et al. (2012), Ryu and Park (2001) regarding various tuning strategies]. Recall now (13), that allows one to write

$$\sigma(t) = \frac{T_{THR}}{C_x}\delta_{THR}(t) + c_1\dot{\phi}(t) + c_0\phi(t) \tag{20}$$

as a function of available feedbacks, ϕ and $\dot{\phi}$, along with δ_{THR}, which is computed by filtering the command signal through the thruster actuator model. Moreover, differentiating (20) with respect to time and using (13) again, one obtains

$$\dot{\sigma}(t) = c_0\dot{\phi}(t) + \frac{T_{THR}}{C_x\tau_{THR}}\left(u_{THR}^{\phi}(t) + \left(c_1\tau_{THR} - 1\right)\delta_{THR}(t)\right). \tag{21}$$

Taking $\dot{\sigma}(t) = -k\,\mathrm{sgn}\,(\sigma(t))$ for a positive tuning constant, $k > 0$, and defining a Lyapunov candidate function $V(t) = \sigma^2(t)$, it follows that $\sigma(t)$ is Lyapunov stable [see Section 3.3 in Khalil (2015)] which will guarantee, via (19), that $\lim_{t\to\infty}\phi(t) = 0$. Then, return to (21) and compute the thruster command as

$$u_{THR}^{\phi}(t) = \left(1 - c_1\tau_{THR}\right)\delta_{THR}(t) - \frac{C_x\tau_{THR}}{T_{THR}}\left(c_0\dot{\phi}(t) + k\,\mathrm{sgn}\,(\sigma(t))\right). \tag{22}$$

Finally, to convert $u_{THR}^{\phi}(t)$ into a usable signal by the cold-gas thrusters (namely u_ϕ, see Fig. 4), a PWPF modulator (see Fig. 7) is employed. Each of the two relays operates one of the "on" and "reverse" settings, while the dead-zone centered in 0 ensures the thrusters are set to "off" when the command signal is negligible and desensitizes the AOCS to measurement noise, which has a tendency to cause needless actuation (chattering). Pitch and yaw AOCS implementations for payload insertion follow identically, by first adapting (13) for these axes.

Remark 16. Recall from (6) that the value of C_x from (19) and (22) is uncertain and deviates from the nominal value that is being gain-scheduled, in accordance with Remark 10. However, rigorous Monte-Carlo simulations on the roll-channel have shown that the Lyapunov stability of the sliding function is in no way compromised and the model's uncertainty has no significant impact upon performance. Moreover, by replacing $\mathrm{sgn}(\sigma(t))$ in (22) with $\frac{\sigma(t)}{|\sigma(t)|+\varepsilon}$ for some $0 < \varepsilon \ll 1$, chattering is reduced at the cost of a negligible decrease in performance.

4. TVC controller validation

4.1. Robust stability analysis

The fundamental requirement that must be fulfilled by the two TVC controllers on the pitch and yaw axes is that the closed-loop system on these channels must be robustly stable at all times during the ascent. To check this, define first

$$\mathbf{G}_\gamma^k = \left[\begin{array}{c|c} 0 & \mathbf{W}_\delta^k\mathbf{G}_{TVC} \\ 0 & \mathbf{W}_{TVC} \\ \hline -\mathbf{G}_{11}^k & -\mathbf{G}_{11}^k\mathbf{G}_{TVC} \\ \mathbf{G}_{21}^l & \mathbf{G}_{21}^k\mathbf{G}_{TVC} \end{array}\right],$$

where the superscript $(\cdot)^k$ denotes that the transfer function under consideration was derived based upon the kth considered instant of the micro-launcher. Split these instants into $k_j \in 1:N_j,\ j \in 1:3$, where j refers to one of the micro-launcher's three stages, and let $\mathbf{K}_j(s)$ denote the controller of the jth stage. Then, by defining

$$\gamma_{k_j}^{stab} = \left\|\mathcal{F}_\ell(\mathbf{G}_\gamma^{k_j}, \mathbf{K}_j)\right\|_\infty,$$

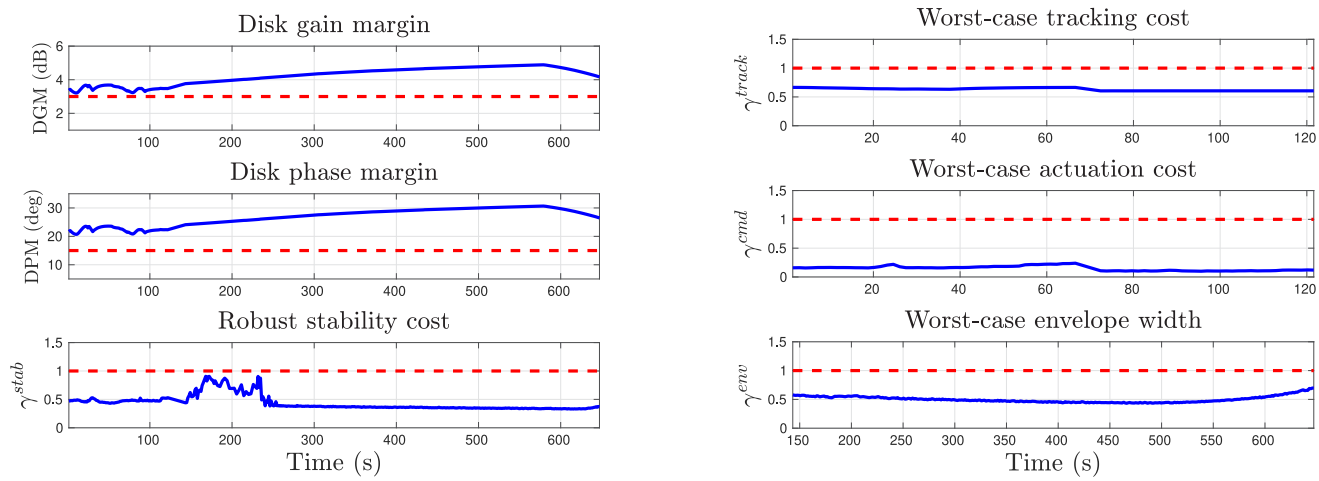
a sufficient condition for robust stability, with respect to the class of systems modeled as in Section 2.4, along the entire trajectory is that $\gamma_{k_j}^{stab} < 1,\ \forall j \in 1:3,\ \forall k_j \in 1:N_j$.

Apart from the above condition, classical specifications for robust control dictate that the proposed control strategy should ensure, for the nominal model of the launch vehicle, a ±6 dB gain margin and a 30 deg phase margin. Note, however, that the employed control loops are not single-input single-output and therefore classical margins fail to apply in this case. To address this, robust stability is also considered with respect to the disk margins, which constitute the largest gain and phase variations that can occur simultaneously both at plant input and at plant output for which the closed-loop system remains stable. Thus, the classical margins are equated with a ±3 dB disk gain margin and a ±15 deg disk phase margin, assigning equal importance to input and output variations.

Remark 17. It becomes clear that the value of $k_\theta^{min} = \frac{1}{\sqrt{2}} \equiv -3$ dB, presented in Remark 15, is in accordance with the imposed disk gain margin. Moreover, once the aforementioned disk margins are ensured, the verification and validation procedure discussed in Section 4 of Navarro-Tapia et al. (2021) can be employed, to assess the control scheme's stability in the presence of flexible body dynamics.

4.2. Worst-case performance analysis

As stated in Section 2.2, the control system must ensure consistent performance regardless of the uncertainty that affects the launch vehicle. Since convex problem statements that allow for the direct design of such a controller are not readily available in the literature, the obtained performance shall be thoroughly validated via a worst-case structured singular value analysis [see Roos and Biannic (2015)].



(a) Robustness indicators (full lines) and bounds (dashed lines)

(b) Performance indicators (full lines) and bounds (dashed lines)

Fig. 8. Controller validation metrics.

Remark 18. Typically, the degradation of performance with respect to plant uncertainty is analyzed via extensive Monte-Carlo simulation suites. Although effective, these are oftentimes both time-consuming and computationally demanding while, due to the stochastic approach in which the sets of parameters are generated, these simulations may also omit critical combinations of uncertain terms which can fail to point out a significant degradation in performance. Thus, the structured singular value approach has been employed, which rapidly offers guarantees for minimum closed-loop performance under mapped uncertainty [see also Menon et al. (2009) for an alternative strategy].

Begin with the first and second stages and recall conditions (C_4) and (C_5) from Section 3.2, which express nominal performance. Then, by (8) and (12), consider $\mathbf{T}^{k_j}_{e_w\theta_r}(\boldsymbol{\Delta}_1,\boldsymbol{\Delta}_3)$ and $\mathbf{T}^{k_j}_{\delta_w\theta_r}(\boldsymbol{\Delta}_1,\boldsymbol{\Delta}_3)$, defined as

$$\mathbf{T}^{k_j}_{e_w\theta_r} = \mathbf{W}_e\left(1-\begin{bmatrix}1 & 0\end{bmatrix}\left(I+\mathbf{G}^{\Delta}_{k_j}(\boldsymbol{\Delta}_1)\begin{bmatrix}1\\0\end{bmatrix}\mathbf{G}^{\Delta}_{TVC}(\boldsymbol{\Delta}_3)\mathbf{K}_j\begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}\right)^{-1} \times\ \mathbf{G}^{\Delta}_{k_j}(\boldsymbol{\Delta}_1)\begin{bmatrix}1\\0\end{bmatrix}\mathbf{G}^{\Delta}_{TVC}(\boldsymbol{\Delta}_3)\mathbf{K}_j\begin{bmatrix}1\\0\end{bmatrix}\right), \tag{23}$$

$$\mathbf{T}^{k_j}_{\delta_w\theta_r} = \mathbf{W}_{cmd}\mathbf{K}_j\left(I-\begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}\left(I+\mathbf{G}^{\Delta}_{k_j}(\boldsymbol{\Delta}_1)\begin{bmatrix}1\\0\end{bmatrix}\mathbf{G}^{\Delta}_{TVC}(\boldsymbol{\Delta}_3)\mathbf{K}_j\begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}\right)^{-1} \times\ \mathbf{G}^{\Delta}_{k_j}(\boldsymbol{\Delta}_1)\begin{bmatrix}1\\0\end{bmatrix}\mathbf{G}^{\Delta}_{TVC}(\boldsymbol{\Delta}_3)\mathbf{K}_j\begin{bmatrix}1\\0\end{bmatrix}\right), \tag{24}$$

by computing the above transfers based upon the control scheme depicted in Fig. 5.

Thus, consider the quantities

$$\gamma^{track}_{k_j} = \sup_{\|\boldsymbol{\Delta}_1\|_\infty,\|\boldsymbol{\Delta}_3\|_\infty\le 1}\left\|\mathbf{T}^{k_j}_{e_w\theta_r}\right\|_\infty, \quad \gamma^{cmd}_{k_j} = \sup_{\|\boldsymbol{\Delta}_1\|_\infty,\|\boldsymbol{\Delta}_3\|_\infty\le 1}\left\|\mathbf{T}^{k_j}_{\delta_w\theta_r}\right\|_\infty;$$

a sufficient condition for robust performance during the first two stages is $\gamma^{track}_{k_j},\gamma^{cmd}_{k_j}<1,\ \forall j\in 1:2,\ \forall k_j\in 1:N_j$.

For the third stage, robust performance is assessed in terms of the inverse stability margin from (16) [see Section 3 of McFarlane and Glover (1992) for more details]. Typically, the open-loop singular value envelope will not greatly deviate from the target loopshape given in (15) if $b(\mathbf{G}_s,\mathbf{K}_s)^{-1}\lessapprox 3$ (recall (16)), and this value is relaxed to a maximum of 6, to account for model singularity when nearing fuel depletion.

As such, define the indicator

$$\gamma^{env}_{k_3} = \sup_{\|\boldsymbol{\Delta}_1\|_\infty,\|\boldsymbol{\Delta}_3\|_\infty\le 1}\frac{1}{6}\left\|\begin{bmatrix}I\\\mathbf{K}_s\end{bmatrix}(I-\mathbf{G}^{k_3}_s\mathbf{K}_s)^{-1}\begin{bmatrix}I & \mathbf{G}^{k_3}_s\end{bmatrix}\right\|_\infty,$$

which was normalized by the maximum deviation and where

$$\mathbf{G}^{k_3}_s(\boldsymbol{\Delta}_1,\boldsymbol{\Delta}_3) = \mathbf{W}_2\mathbf{G}^{\Delta}_{k_3}(\boldsymbol{\Delta}_1)\mathbf{G}^{\Delta}_{TVC}(\boldsymbol{\Delta}_3)\mathbf{W}_1;$$

a sufficient condition for third stage robust performance is $\gamma^{env}_{k_3}<1,\ \forall k_3\in 1:N_3$.

5. Numerical results and simulations

5.1. Control system validation results

As can be seen in Fig. 8(a), all robust stability criteria with respect to the rigid-body's dynamics (recall also Remark 17) are ensured throughout the ascent and the obtained disk margins, taken in absolute value, are above the lower bounds given in Table 1. Moreover, the worst-case performance analysis, illustrated via Fig. 8(b), shows that, even for the most unfavorable combination of uncertain parameters (recall Section 2.4), the closed-loop system maintains its robust performance.

5.2. Mission simulation results

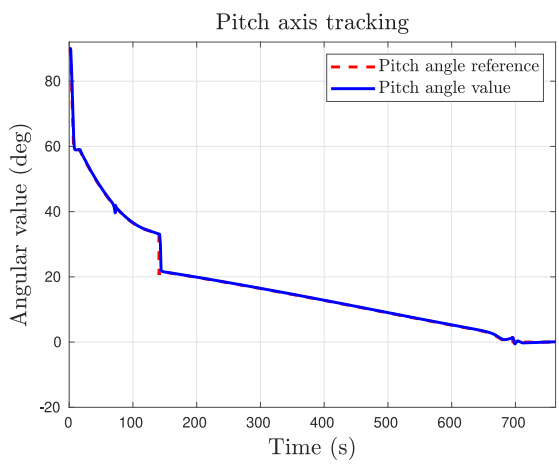
The control system showcased in Section 3.1 has been designed and tested upon the micro-launcher prototype's flight simulator, which implements its nonlinear and uncertain equations of motion, along with flexible bending and atmospheric wind. The simulated mission was successful, with orbit circularization occurring at $t = 695$ s.

Figs. 9(a) and 10(a) display the pitch and yaw angles, which closely track their prescribed references, with the natural exceptions of the two coasting phases, when the rocket lacks actuation (recall the intervals discussed in Section 2.1); Fig. 11(a) shows that the roll angle is kept within ±2 deg, further showcasing the performance achieved by the control scheme.

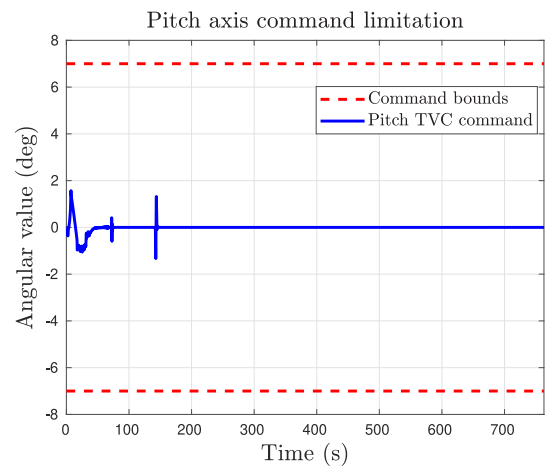
Figs. 9(b) and 10(b) display the computed TVC nozzle deflection angles, which are well within prescribed bounds; Fig. 11(b) shows that roll damping actuation is triggered at 30 second intervals, successfully preserving fuel for payload insertion.

Finally, Figs. 12(a) and 12(b) show that, after the imposed settling times, both the angle of attack and the sideslip angle are kept within their prescribed bounds for the duration of the entire gravity-turn maneuver, both before first stage separation ($t = 68$ s) and after the second stage's activation ($t = 71$ s).

Note that the aforementioned bounds were deduced with respect to the maximum acceptable value of the product between the aerodynamic incidence angle (depending on the considered motion axis) and the dynamic pressure [see Section 4.2 of Valli et al. (2015) for a detailed explanation].

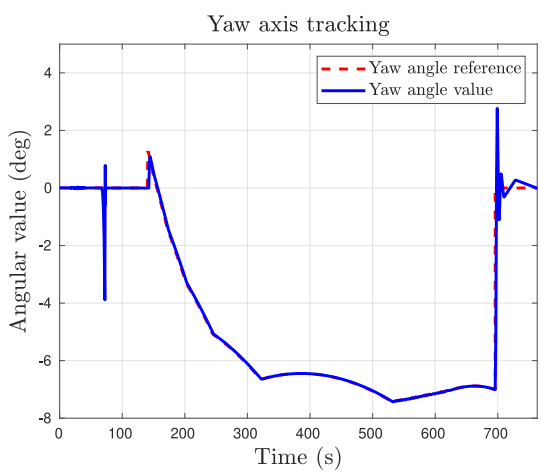


(a) Pitch angle reference is successfully tracked

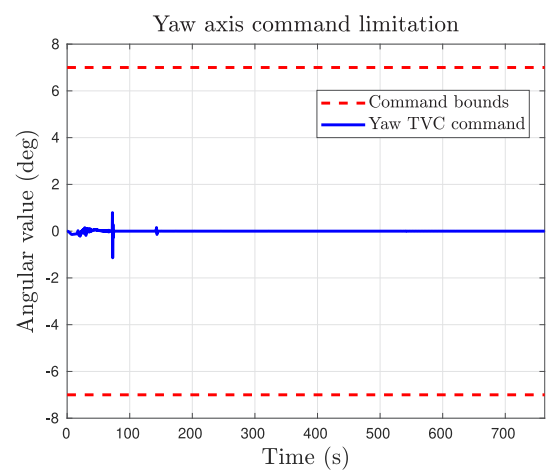


(b) Pitch axis command is kept within implementable bounds

Fig. 9. Pitch axis performance analysis.

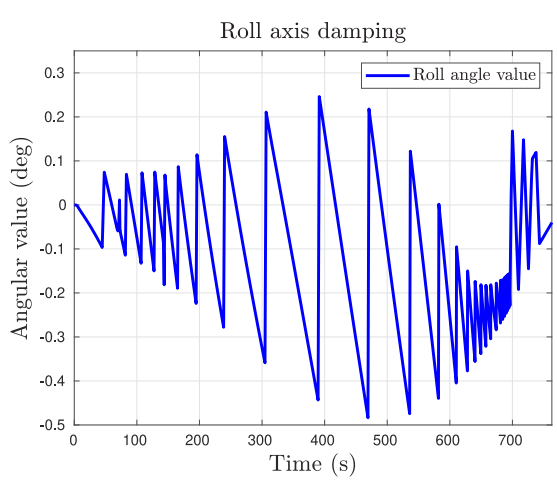


(a) Yaw angle reference is successfully tracked

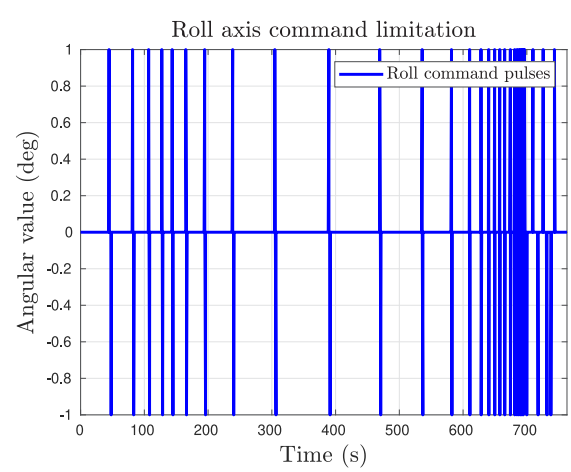


(b) Yaw axis command is kept within implementable bounds

Fig. 10. Yaw axis performance analysis.

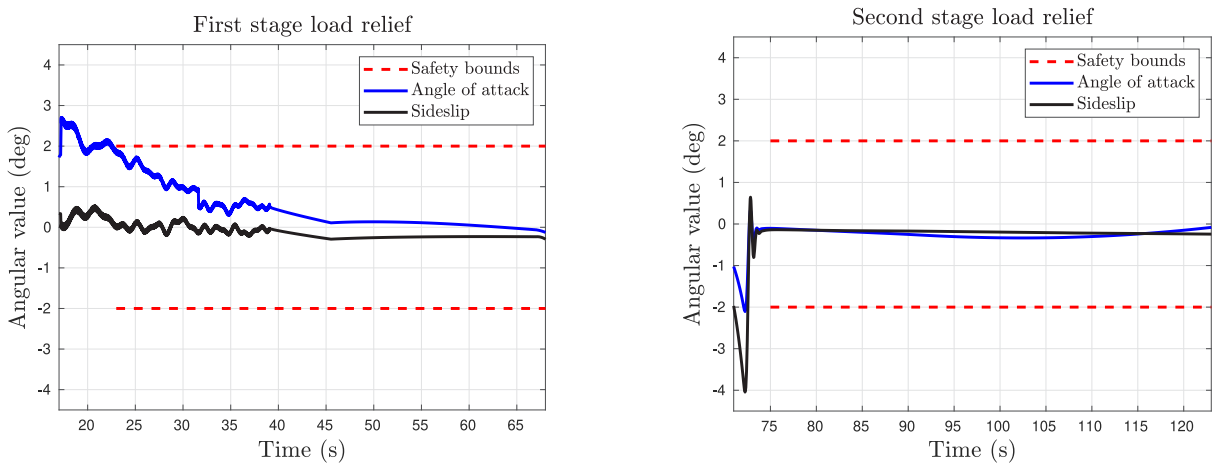


(a) Roll axis command is triggered infrequently to preserve fuel



(b) Roll angle is successfully kept small

Fig. 11. Roll axis performance analysis.



(a) Pre-coasting gravity turn occurs safely

(b) Post-coasting gravity turn occurs safely

Fig. 12. Angle of attack and sideslip angle minimization during gravity turn.

Since Figs. 9(a) and 10(a) showcase good reference tracking during gravity turn, load relief is achieved by following a trajectory which minimizes both the angle of attack and the sideslip angle, respectively.

6. Final comments and further developments

6.1. Conclusions

For a satellite micro-launcher prototype subject to parametric rigid-body model uncertainty and flexible-body bending, a complete control scheme is proposed, whose chief components tackle the individual challenges that make up the flight mission. These control subroutines are designed to complement each other's functions and are then rigorously tested to ensure consistent performance in the face of model uncertainty and time-varying disturbance, such as wind interference and bending mode oscillations. The orbital insertion mission is simulated upon the micro-launcher's nonlinear flight simulator and the mission's success validates the proposed control solution.

6.2. Future objectives

One of the aspects which would benefit from additional attention is the sensitivity of the optimization problem in (9) to the numerical conditioning of the micro-launcher's realization. As stated in Section 3.3, the third stage is particularly badly conditioned, due to the poles in the origin of the complex plane, and the result of these singularities can be seen just after the 140 second mark in Fig. 8(a), when the robust stability cost rises due to the conservative estimates of the uncertainty and the maximum level of achievable performance may decrease. In order to tackle the aforementioned conservatism, another viable approach would be to select an acceptable degree of consumer risk [type II error, see Hanson and Beard (2012)], rather than the thousand sample-based Monte Carlo strategy presented in Section 2.4.

Moreover, note that the adaptive augmentation scheme has been utilized for bending mode compensation without having to redesign the TVC controllers. If properly tuned, the adaptive scheme could also be employed to account for changes in payload mass and reuse the entire control subroutine for multiple missions with different payloads by using the improved, frequency-weighted adaptive scheme introduced in Navarro-Tapia et al. (2021) to counteract the effects of these changes in mass and its distribution.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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